StarDust — One-Page Quick Revision

One line: Monitoring retrofit for *existing* drain trash booms: Δh across the boom + one mooring-line tension → ESP32 validity checks + baseline → NORMAL / WATCH / INSPECTION REQUEST / UNCERTAIN / FAULT → human crew.

Problem: Booms intercept nullah plastic (≈5,000 t/yr estimate in brief; BMC ₹14.5 cr 2-yr O&M contract, FPJ 30 Jul 2026). Condition between scheduled visits is unknown.

Contribution (not the sensors): site-calibrated combined decision logic with explicit fault/uncertainty, *tested* against level-only and tension-only.

Maths: $h = z - d$ · $\Delta h = hA - hB$ · $T = k(raw - offset)$ [N] · uncertainty ≈ $\sqrt{(uA^2+uB^2)}$

Logic: valid? → in tested regime? → baseline + persistence. Independent valid high-water/high-load warnings never suppressed.

Test: rig A clean · B debris · C flow change · D backwater · E fault+recovery. 60 held-out runs. FPR, miss rate, delay, uncertain rate, valid-data fraction. Gates: measurable → beats simpler → fault-safe.

Cost: ₹8,650 + 15% = ₹9,947.50 ≈ **₹9,950** bench planning estimate. Not field price.

Never say: results · accuracy · prevents floods · world-first · BMC partner · measures plastic weight · site confirmed.

Hardest Qs: What's new? · Debris vs flow? · Where are results? · What if combined doesn't help? (→ use simpler; still useful) · Why pay vs manual? (multi-site owner, unverified) · Site confirmed? (no, Malad candidate) · Flood prevention? (no) · Environmental attribution? (independent weighing vs baseline)

Limitations: tide/backwater · small Δh on shallow booms · one line ≠ structure · bench ≠ monsoon · needs a crew.